

Penetration Test Report

Prepared for Hotel Dorsey



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Introduction

This report summarizes the results of an authorized penetration test conducted for Hotel Dorsey. The purpose of this assessment was to determine whether vulnerabilities discovered during the initial security scan could be exploited by a malicious attacker to gain unauthorized access to the system and extract sensitive data.

All testing activities were conducted within the defined scope and authorization provided by the client. The attack simulation was performed from a controlled testing environment using Kali Linux, a widely used security testing operating system designed for penetration testing and vulnerability assessment.

Several industry standard tools were used during this engagement. These included Zenmap, which is a graphical interface for the Nmap network scanner used to identify open ports and services on a system. Metasploit and other security tools were used to attempt exploitation of vulnerabilities discovered during the scanning phase. Password auditing tools such as John the Ripper were also used to demonstrate how weak or exposed credentials could be cracked by an attacker.

The objective of this penetration test was to determine whether vulnerabilities present on the system could allow unauthorized access, credential theft, and potential data exfiltration that could impact the organization's confidentiality, integrity, and availability of information.

Target

The penetration test focused on a Linux server belonging to Hotel Dorsey. The system was scanned to identify open ports and running services that could potentially be exploited by an attacker.

Target IP Address	10.6.3.50
Attack Machine IP Address	10.6.3.100

Figure 1: Zenmap Scan Results

Hosts	Services	Nmap Output	Ports / Hosts	Topology	Host Details	Scans
OS	Host	Port	Protocol	State	Service	Version
10.6.3.100		21	tcp	open	ftp	vsftpd 2.3.4
		22	tcp	open	ssh	OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
		23	tcp	open	telnet	Linux telnetd
		25	tcp	open	smtp	Postfix smtpd
		53	tcp	open	domain	ISC BIND 9.4.2
		80	tcp	open	http	Apache httpd 2.2.8 (Ubuntu DAV/2)
		111	tcp	open	rpcbind	2 (RPC #100000)
		139	tcp	open	netbios-ssn	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
		445	tcp	open	netbios-ssn	Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
		512	tcp	open	exec	netkit-rsh rshexec
		513	tcp	open	login	OpenBSD or Solaris rlogind
		514	tcp	open	shell	Netkit rshd
		1099	tcp	open	java-rmi	Java RMI Registry
		1524	tcp	open	bindshell	Bash shell (**BACKDOOR** root shell)
		2049	tcp	open	nfs	2-4 (RPC #100003)
		2121	tcp	open	rtp	ProFTPD 1.3.1
		3306	tcp	open	mysql	MySQL 5.0.51a-3ubuntu5
		5432	tcp	open	postgresql	PostgreSQL DB 8.3.0 - 8.3.7
		5901	tcp	open	vnc	VNC (protocol 3.3)
		6001	tcp	open	x11	(access denied)
		6667	tcp	open	irc	UnrealIRCd
		8009	tcp	open	alpl3	Apache Jserv (Protocol v1.3)
		8180	tcp	open	http	Apache Tomcat/Coyote JSP engine 1.1

Vulnerability

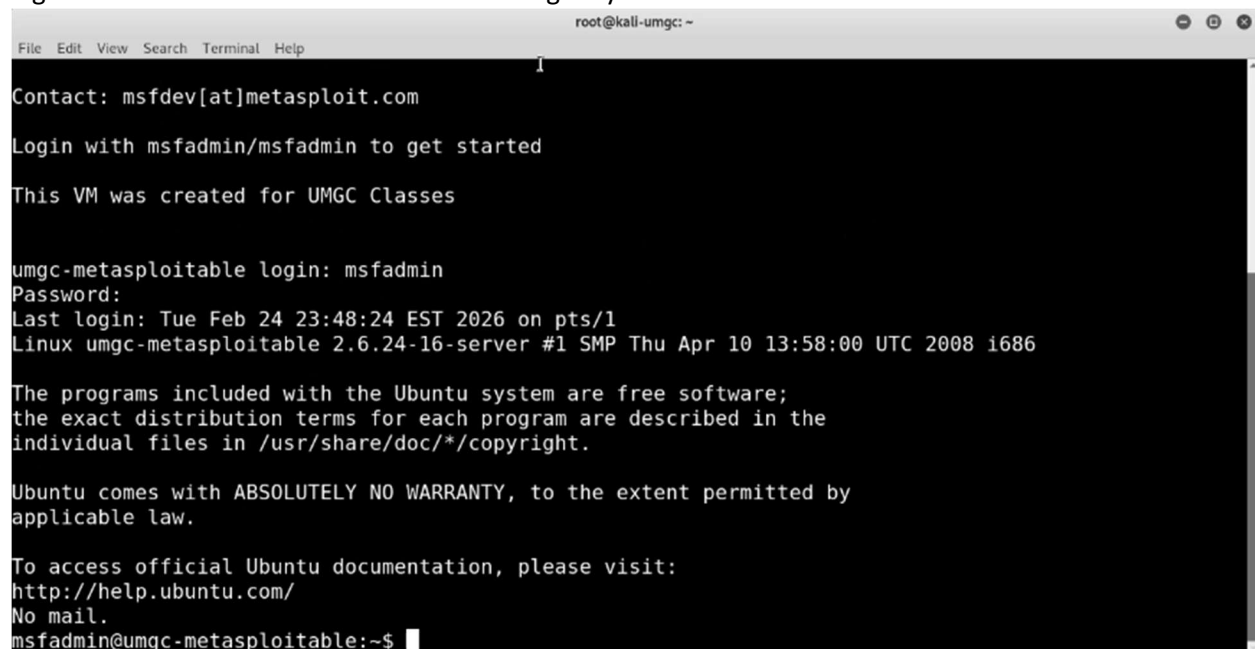
During the penetration test, a critical vulnerability was identified that allowed remote command access to the system through port 1524. This port provided an unauthorized shell connection, which allowed commands to be executed directly on the target system.

Using Telnet, the penetration tester was able to connect to the system and gain access to a command shell. This type of vulnerability is extremely dangerous because it allows attackers to bypass authentication mechanisms and interact directly with the operating system.

Once connected to the system, commands were executed to verify the level of access obtained. The whoami and getuid commands confirmed that access had been successfully established on the system.

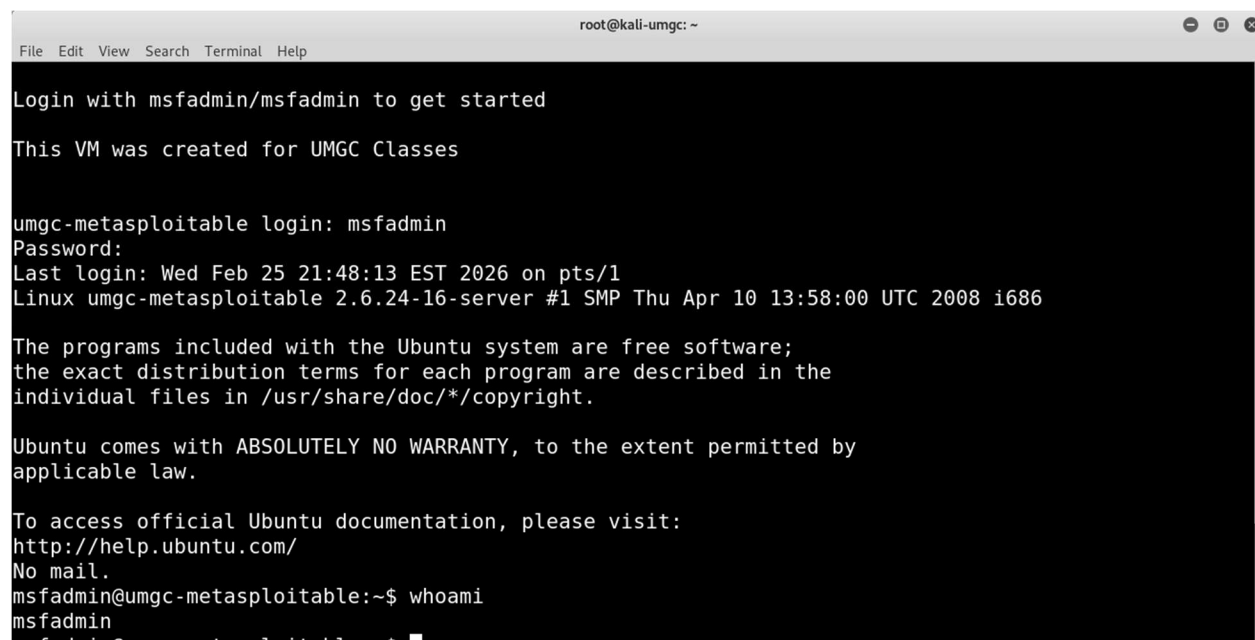
This demonstrated that an attacker could remotely execute commands and interact with files stored on the server.

Figure 2: Successful Telnet Connection to Target System



```
root@kali-umgc: ~  
File Edit View Search Terminal Help  
1  
Contact: msfdev[at]metasploit.com  
Login with msfadmin/msfadmin to get started  
This VM was created for UMGC Classes  
  
umgc-metasploitable login: msfadmin  
Password:  
Last login: Tue Feb 24 23:48:24 EST 2026 on pts/1  
Linux umgc-metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To access official Ubuntu documentation, please visit:  
http://help.ubuntu.com/  
No mail.  
msfadmin@umgc-metasploitable:~$
```

Figure 3: Access Level (whoami Command Output)



```
root@kali-umgc: ~  
File Edit View Search Terminal Help  
Login with msfadmin/msfadmin to get started  
This VM was created for UMGC Classes  
  
umgc-metasploitable login: msfadmin  
Password:  
Last login: Wed Feb 25 21:48:13 EST 2026 on pts/1  
Linux umgc-metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To access official Ubuntu documentation, please visit:  
http://help.ubuntu.com/  
No mail.  
msfadmin@umgc-metasploitable:~$ whoami  
msfadmin  
msfadmin@umgc-metasploitable:~$
```

The ability to obtain command line access significantly increases the risk of further compromise. From this position, attackers may attempt to privilege escalation, install malicious software, or access sensitive company data.

Data Exfiltration

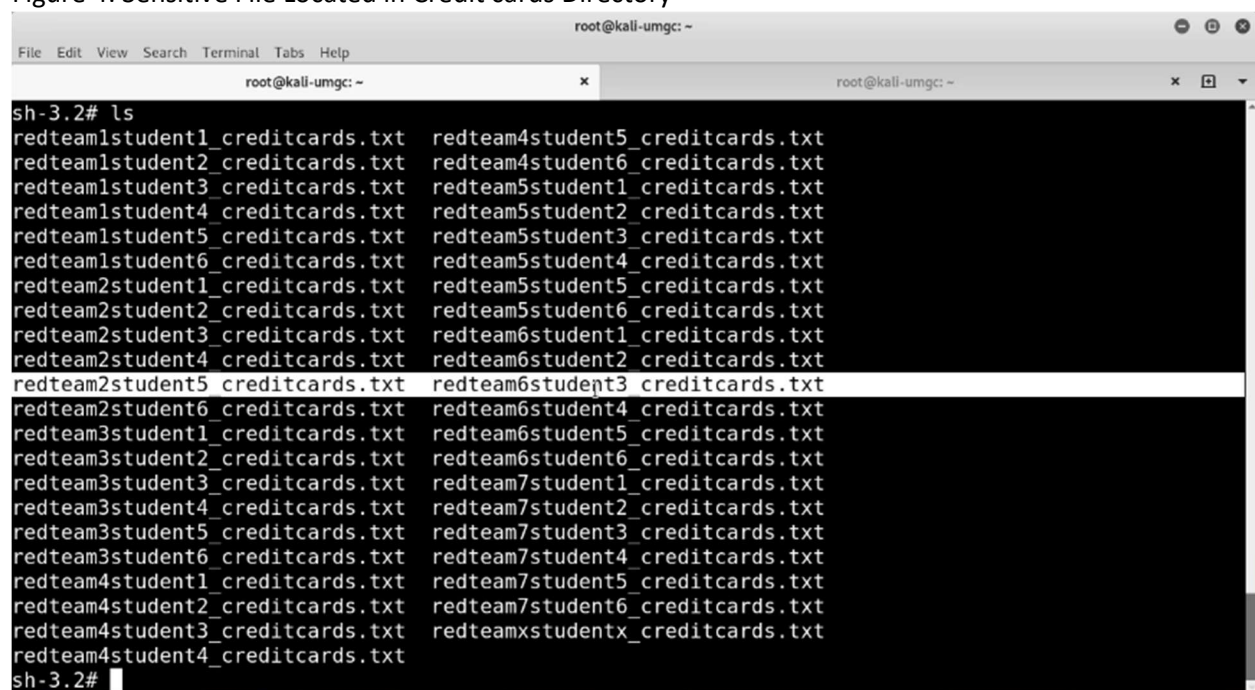
After gaining access to the system, the penetration tester demonstrated how sensitive information could be extracted from the network.

While browsing the system directories, several folders containing sensitive files were discovered. One example included a directory containing confidential financial information.

To demonstrate data exfiltration, a file located within the credit cards directory was copied to a publicly accessible FTP directory. This allowed the file to be accessed externally through the web server.

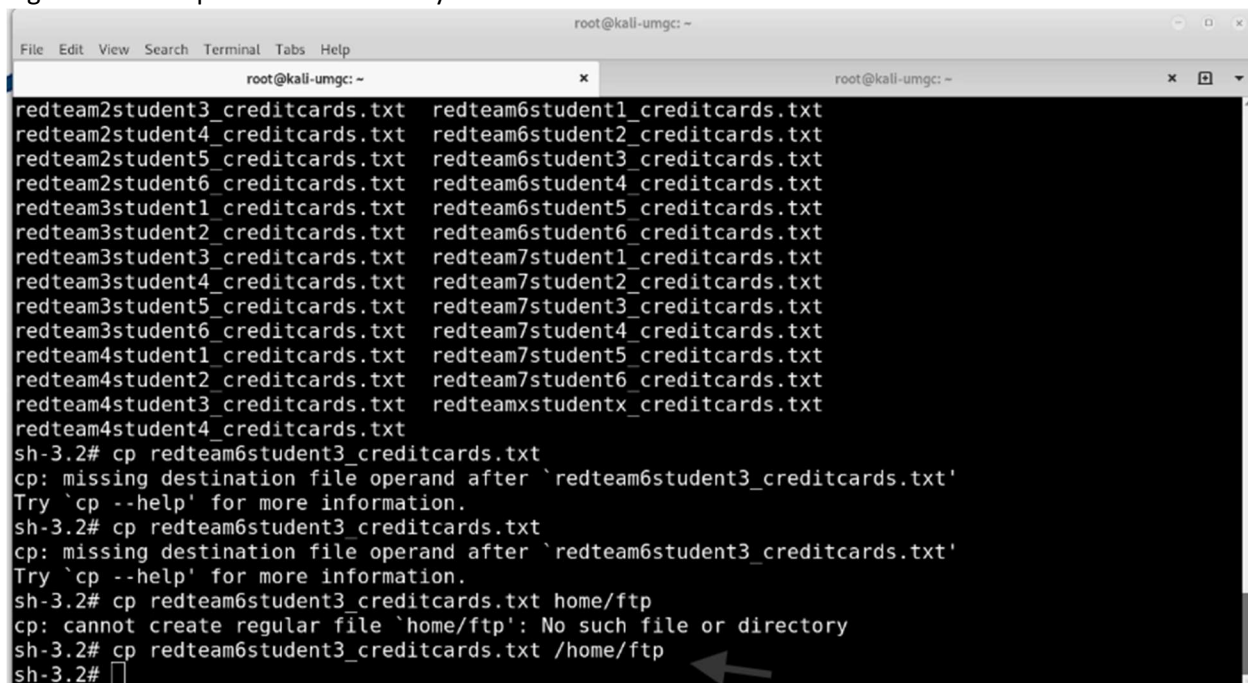
This technique illustrates how attackers can move sensitive data to a location where it can be easily retrieved without triggering traditional security alerts.

Figure 4: Sensitive File Located in Credit cards Directory



```
root@kali-umgc: ~  
File Edit View Search Terminal Tabs Help  
root@kali-umgc: ~  
sh-3.2# ls  
redteam1student1_creditcards.txt redteam4student5_creditcards.txt  
redteam1student2_creditcards.txt redteam4student6_creditcards.txt  
redteam1student3_creditcards.txt redteam5student1_creditcards.txt  
redteam1student4_creditcards.txt redteam5student2_creditcards.txt  
redteam1student5_creditcards.txt redteam5student3_creditcards.txt  
redteam1student6_creditcards.txt redteam5student4_creditcards.txt  
redteam2student1_creditcards.txt redteam5student5_creditcards.txt  
redteam2student2_creditcards.txt redteam5student6_creditcards.txt  
redteam2student3_creditcards.txt redteam6student1_creditcards.txt  
redteam2student4_creditcards.txt redteam6student2_creditcards.txt  
redteam2student5_creditcards.txt redteam6student3_creditcards.txt  
redteam2student6_creditcards.txt redteam6student4_creditcards.txt  
redteam3student1_creditcards.txt redteam6student5_creditcards.txt  
redteam3student2_creditcards.txt redteam6student6_creditcards.txt  
redteam3student3_creditcards.txt redteam7student1_creditcards.txt  
redteam3student4_creditcards.txt redteam7student2_creditcards.txt  
redteam3student5_creditcards.txt redteam7student3_creditcards.txt  
redteam3student6_creditcards.txt redteam7student4_creditcards.txt  
redteam4student1_creditcards.txt redteam7student5_creditcards.txt  
redteam4student2_creditcards.txt redteam7student6_creditcards.txt  
redteam4student3_creditcards.txt redteamxstudentx_creditcards.txt  
redteam4student4_creditcards.txt  
sh-3.2#
```

Figure 5: File Copied to FTP Directory

A screenshot of a terminal window titled 'root@kali-umgc: ~'. The terminal shows a list of files in the current directory, including various 'creditcards.txt' files. Below the list, several 'cp' commands are entered. The first two fail with 'cp: missing destination file operand after'. The third fails with 'cp: cannot create regular file 'home/ftp': No such file or directory'. The fourth command is 'cp redteam6student3_creditcards.txt /home/ftp', and the cursor is at the end of the line. A white arrow points to the end of the last command line.

```
root@kali-umgc: ~
File Edit View Search Terminal Tabs Help

redteam2student3_creditcards.txt redteam6student1_creditcards.txt
redteam2student4_creditcards.txt redteam6student2_creditcards.txt
redteam2student5_creditcards.txt redteam6student3_creditcards.txt
redteam2student6_creditcards.txt redteam6student4_creditcards.txt
redteam3student1_creditcards.txt redteam6student5_creditcards.txt
redteam3student2_creditcards.txt redteam6student6_creditcards.txt
redteam3student3_creditcards.txt redteam7student1_creditcards.txt
redteam3student4_creditcards.txt redteam7student2_creditcards.txt
redteam3student5_creditcards.txt redteam7student3_creditcards.txt
redteam3student6_creditcards.txt redteam7student4_creditcards.txt
redteam4student1_creditcards.txt redteam7student5_creditcards.txt
redteam4student2_creditcards.txt redteam7student6_creditcards.txt
redteam4student3_creditcards.txt redteamxstudentx_creditcards.txt
redteam4student4_creditcards.txt

sh-3.2# cp redteam6student3_creditcards.txt
cp: missing destination file operand after `redteam6student3_creditcards.txt'
Try `cp --help' for more information.
sh-3.2# cp redteam6student3_creditcards.txt
cp: missing destination file operand after `redteam6student3_creditcards.txt'
Try `cp --help' for more information.
sh-3.2# cp redteam6student3_creditcards.txt home/ftp
cp: cannot create regular file `home/ftp': No such file or directory
sh-3.2# cp redteam6student3_creditcards.txt /home/ftp
sh-3.2#
```

The impact of this type of breach could be significant for Hotel Dorsey. Exposure of financial data or customer information could lead to:

- Loss of customer trust
- Legal liability and regulatory penalties
- Financial damages from fraud
- Reputation damage and potential stock value decline

This demonstration confirms that the vulnerabilities identified during the scan phase are not theoretical and could be actively exploited by attackers.

Recommendations

To mitigate the vulnerabilities identified during this penetration test, several security improvements are recommended.

First, the organization should immediately disable or restrict access to unnecessary services, particularly Telnet and any unauthorized backdoor services such as port 1524. Telnet should be replaced with Secure Shell (SSH), which provides encrypted remote access.

Second, the system should undergo patching and system hardening to ensure that outdated services and known vulnerabilities are removed. Keeping systems updated is critical for preventing exploitation.

Next, the organization should implement strong authentication and password policies. Weak credentials can be easily cracked using password auditing tools such as John the Ripper. The network segmentation

and firewall rules should be configured to restrict external access to sensitive services. Only required services should be exposed to the network.

Finally, the organization should implement continuous vulnerability scanning and monitoring tools to identify security issues before attackers can exploit them. By implementing these controls, Hotel Dorsey can significantly reduce the risk of unauthorized access and protect sensitive company information.

References

Nmap Network Scanning Guide

<https://nmap.org/book>

Metasploit Framework Documentation

<https://docs.metasploit.com>

Kali Linux Penetration Testing Platform

<https://www.kali.org>

John the Ripper Password Cracker

<https://www.openwall.com/john>